Ajay Venkatesh

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Education

New York University, New York, NY — *Aug 2024 – May 2026*

Master of Science in Computer Engineering (GPA: 3.9/4.0)

Coursework: Machine Learning, Deep Learning, Big Data, Computer Architecture, System Design, Data Structures

Experience

- **Campus Mesh** — *Jan 2026 – Present*
  AI Software Engineer — New York, US
  - Built **CampusIQ**, an LLM-powered AI assistant, with iterative **prompt engineering** and **RAG retrieval** over structured data and transformer-based embeddings, improving grounded accuracy by **30%**.
  - Designed a personalized **recommendation engine** using collaborative filtering on behavioral signals, increasing engagement by **28%** and retention by **50%**.
  - Developed scalable **Node.js microservices** powering real-time academic workflows for **10K+ users**.
- **Amazon** — *May 2025 – Aug 2025*
  Software Development Engineer Intern — Seattle, US
  - Shipped a **RAG pipeline** with an **LLM agent workflow** and **function calling** that autonomously retrieves operational logs, reasons over incident details, and generates grounded remediation steps.
  - Built a **Model Context Protocol (MCP)** server for tool-augmented retrieval, cutting AWS Athena query time and compute costs by **40%** via better LLM inference efficiency.
  - Built a high-throughput, event-driven backend on **AWS ECS Fargate, SQS, SNS** with **Infrastructure-as-Code** via **AWS CDK**, decoupling distributed message processing and ensuring **99.9%** availability under 2x peak load.
  - Engineered the backend services in **Java** (Coral, Smithy, Dagger, CBOR); applied **low-level design patterns**, cutting execution time by **35%** with unit/integration testing via Mockito.
- **New York University (Novel AI Technologies, Inc.)** — *Aug 2024 – Present*
  AI Software Engineer — New York, US
  - Architected an **NSF-funded ($100K)** AI-powered health platform for **iOS** and **Android** ingesting real-time wearable telemetry; implemented **time-series anomaly detection** using **Isolation Forest** and **LSTM** models.
  - Built end-to-end **computer vision pipelines** (**YOLOv8, EfficientNet**) for user-submitted food images, integrated with **GPT-4 vision workflows** for nutritional estimation and dietary feedback.
  - Developed an insurance analytics platform (**React, Node.js, REST APIs**) integrating **LLM-based document analysis** (**GPT-4 + LangChain**) for underwriting insights.
  - Engineered **ETL pipelines** transforming **NoSQL** into **PostgreSQL** with concurrency control; processed **250K+** records and powered **AWS QuickSight** dashboards.
- **PricewaterhouseCoopers (PwC)** — *Aug 2021 – Aug 2024*
  Senior Software Engineer — Bengaluru, India
  - Designed a distributed backend on **Microsoft Azure** integrating with Microsoft Dynamics and supporting **ML inference pipelines** for healthcare analytics with API-based integrations.
  - Built an **NLP-powered OCR pipeline** using **LayoutLM transformer** and Azure Form Recognizer to automate structured data extraction from documents, reducing processing time by **75%**.
  - Optimized data ingestion with **parallel processing** on **100K+** records, cutting execution from **5 hours to 1 hour**.
  - Led code reviews and mentored engineers on **software design and test-driven development**, reducing production bugs by **25%**.

Projects

- **RV32I Processor Simulator** | *Python, RISC-V ISA, Computer Architecture* [GH]
  - Built a cycle-accurate **RV32I simulator** in **Python** with single-stage and **5-stage pipelined (IF/ID/EX/MEM/WB)** cores, implementing R-type, I-type, LW/SW, BEQ/BNE, JAL with two's-complement immediate handling and opcode-based dispatch.
  - Engineered **hazard detection** with **EX/MEM-to-EX forwarding** and load-use stalls, resolving RAW dependencies without state rollback.
  - Generated per-cycle register/memory/pipeline dumps and computed **CPI & IPC** metrics, validated across multiple assembly test cases.
- **Flight Delay Prediction (Big Data)** | *Hadoop HDFS, PySpark, Spark SQL, Spark MLlib, Seaborn, Plotly* [GH]
  - Built a distributed big data pipeline over **Hadoop HDFS** on **2024 BTS US flight data**; used **PySpark** and **Spark SQL** for cleaning, feature engineering, and large-scale aggregations.
  - Trained **Spark MLlib** classification models for delay severity prediction with feature-importance analysis via Seaborn, Matplotlib, and Plotly.
- **InviLite** | *AWS, SageMaker, Bedrock, FAISS, Python*
  - Architected a cloud-native data pipeline (Kinesis, Lambda, S3 Bronze/Silver/Gold, Glue, DynamoDB) ingesting structured and unstructured enterprise data into governed ML pipelines.
  - Built a **RAG system** (SVM classification + FAISS vector indexing + **Bedrock LLM**), cutting incident resolution from **1–2 hours to minutes** and lifting root-cause classification accuracy from **56% to 92%** via feature engineering and model retraining.

Skills

- **Programming Languages:** Python, C++, Java, Bash, SQL
- **AI/ML:** PyTorch, TensorFlow, CUDA, Transformers, RAG (LangChain, FAISS), LLM Fine-tuning, LLM Agents, Reinforcement Learning, Deep Learning, NLP, Hugging Face, Pandas, NumPy, MLflow
- **Big Data & Distributed:** Apache Spark, PySpark, Spark SQL, Spark MLlib, Hadoop HDFS, ETL/ELT Workflows
- **Cloud & DevOps:** AWS (Lambda, SageMaker, Bedrock, ECS, S3, CloudWatch, CDK), Azure (AI Services, Data Lake, Function Apps), Docker, Kubernetes, CI/CD, Terraform
- **Tools & Platforms:** Linux, Git, Jupyter, Cursor, GitHub Copilot, Model Context Protocol (MCP), RISC-V

Publications

Supervised ML System Based Segmentation and Classification of Stroke Using Deep Learning Techniques *(IEEE)*

An Aquila-optimized SVM Classifier for Diabetes Prediction *(IEEE)*

Certifications

AWS Cloud Practitioner | Stanford Machine Learning (Coursera) | Azure Data Fundamentals